An energy budget

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect work, motion, and heating. State a conclusion that fits the calculation's limits.

SUCCESS CRITERIA - I can...

1. Keep all terms in consistent energy units.
2. Distinguish mechanical-energy change from total-energy loss.
3. State what you treat as true when you account for the energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An energy account begins with **15 J**. It ends with **9 J** motion energy and **3 J** thermal energy. **Find the unaccounted energy** and suggest a next check.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $E_i + E_{in} = E_f + E_{out}$.
i/f = initial/final stored energy; in/out = transfers across the chosen system boundary. Account for every form.

Worked reasoning: An input of **100 J** gives **60 J** useful motion, **25 J** thermal transfer and **15 J** sound/other transfers. The account balances because **60**+**25**+**15**=**100 J**.

Your turn: Draw proportional energy arrows and **calculate the useful fraction**.

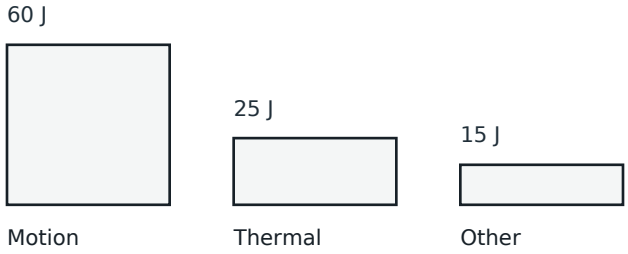
Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A modeled system starts with **100 J**. It ends with **60 J** motion energy, **25 J** thermal energy and **15 J** sound/deformation. **Draw an energy account** and **check the total**.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

04 Apply the physics

A **2 kg** cart starts at **4 m/s** and receives **8 J** work. **Find initial** and final kinetic energy if no losses. Then include **5 J** transferred to thermal energy. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

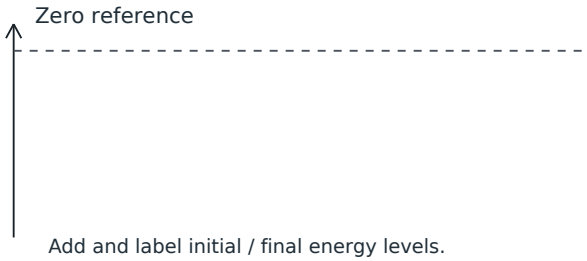
05 Explain your reasoning

What information is missing if an impact report gives only total kinetic energy and says all of it heats the ground?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Gravitational potential energy changes from **-24 J** to **-4 J**. Find ΔU and state whether energy is gained or lost. **Label the distances or energy** levels used in your reasoning.



Add and label initial / final energy levels.

Marzano check - circle: 1 2 3

07 Exit check

Input energy **40 J** yields **30 J** useful mechanical energy.
1. **Find other energy** and useful fraction.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Improve useful output to **75 J** without changing input.
1. Design two possible new thermal/other accounts and **explain why data would be** needed to **choose one**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3